

Name

Date:

Unit 3 Test:

Electric Force, Field, Potential Difference

K	T	A	C
/ 20	/ 20	/ 10	/ 10

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

APPLICATION

1. Three charged particles, are arranged in a line, as shown in the diagram. Calculate the net electrostatic force on Q_3 ($-4.0 \times 10^{-6} \text{ C}$) due to the other two charges. Show all your work.

$$\text{Net } \Sigma F \text{ on } Q_3$$

Same line

$$F_{1 \text{ on } 3} = \frac{k Q_1 Q_3}{0.5^2}$$

$$= 1.15072 \text{ N to the right}$$

$$F_{2 \text{ on } 3} = k \frac{3 \mu\text{C} \cdot 4 \mu\text{C}}{0.2^2}$$

$$F_{2 \text{ on } 3} = 2.6971 \text{ N to the left}$$

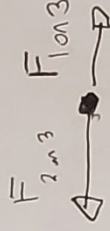
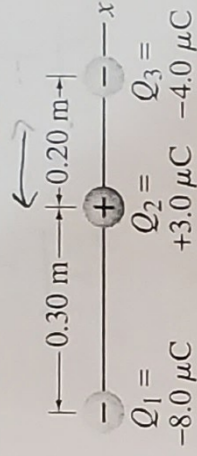
$$1.15072 - 2.6971 = \Sigma F_x$$

$$-1.54638 \text{ N} = \Sigma F_x$$

$$\therefore 1.54 \text{ Newtons of force to the left}$$

$$L_3^+$$

[3.5 marks]

Final answer: 1.55 N left

-

$$X = \frac{1-25}{8}$$

$$x = 2(\sqrt{2} + 1)$$

2

$$\sqrt[3]{-b \pm \sqrt{b^2 - 4ac}}$$

3

$$x = 1 \left(\frac{-2 \pm \sqrt{5}}{-2} \right)$$

241000 - 0.41

25.0 -

$$\frac{-1}{-20 + \sqrt{40^2 + 40^2}}$$

Final answer: $\left(\frac{-2-\sqrt{8}}{-9}\right) d = x$

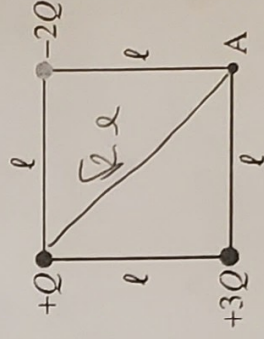
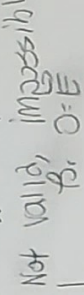
- We sum up potentials

$$\frac{y}{x} = y$$

[3 marks]

$$\frac{-15 \pm \sqrt{510}}{5}$$

$$\frac{1}{\sqrt{2}} \left(\frac{\sqrt{2}}{1+i} \right)$$



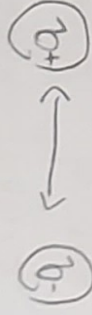
Final answer: $\frac{12}{2} \left(\frac{\sqrt{2+1}}{\sqrt{2}} \right)$ ✓

Will Donel.

THINKING/INQUIRY

1. A charge Q is transferred from an initially uncharged plastic ball to an identical ball 24 cm away. The force of attraction is then 17 mN. How many electrons were transferred from one ball to the other?
Show all your work.

$$0.24 \text{ m} = r$$



[3 marks]

$$F = k \frac{q_1 q_2}{r^2}$$

$$\frac{3.3 \times 10^{-19}}{1.6 \times 10^{-19}} = 2.062$$

$$F = \frac{(Q)(Q)}{r^2}$$

$$17 \times 10^{-3} \text{ N} = k \frac{Q^2}{r^2}$$

so 2.062×10^{12} electrons have been transferred

$$\sqrt{\frac{17 \times 10^{-3} \text{ N} \cdot r^2}{k}} = Q$$

$$3.300 \times 10^{-7} \text{ C} = Q$$

Final answer: 2.062×10^{12} electrons

2. You are given two unknown point charges Q_1 and Q_2 as shown in the diagram, and at a point on the line joining them, one-third of the way from Q_1 to Q_2 , the electric field is zero. What is the ratio from Q_1/Q_2 ?
Show all your work.

$$E + E = 0$$

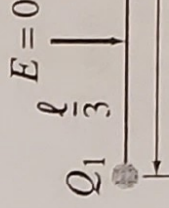
$$\frac{E}{\left(\frac{1}{3}\right)^2} = \frac{E}{\left(\frac{2}{3}\right)^2}$$

$$\frac{kQ_1}{\left(\frac{1}{3}\right)^2} = \frac{kQ_2}{\left(\frac{2}{3}\right)^2}$$

$$\frac{Q_1}{\left(\frac{1}{3}\right)^2} = \frac{Q_2}{\left(\frac{2}{3}\right)^2}$$

$$\frac{4Q_1}{9} = Q_2 \frac{1}{9}$$

$$4Q_1 = Q_2$$



[3 marks]

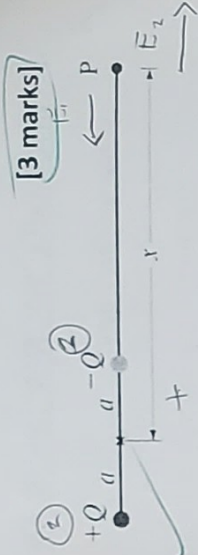
Final answer: $4Q_1 = Q_2$

$$\frac{Q_1}{Q_2} = \frac{1}{4}$$

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3. Determine the direction and magnitude of the electric field at the point P as shown in the diagram. The charges are separated by a distance $2a$, and point P is a distance x from the midpoint between the two charges. Express your answer in terms of Q , x , a , and k .

Field is $kQ \frac{1}{r^2}$

$$\vec{E}_2 = kQ \frac{(x+a)^2 - (x-a)^2}{(x+a)^2(x-a)^2}$$

$$\vec{E}_1 = k \left(\frac{Q}{(x-a)^2} \right)$$

$$= \frac{(x^2 + 2ax + a^2) - (x^2 - 2ax + a^2)}{(x+a)^2(x-a)^2}$$

$$= \frac{4ax}{(x+a)^2(x-a)^2}$$

$$\vec{E} = k \frac{Q}{(x+a)^2} - k \frac{Q}{(x-a)^2}$$

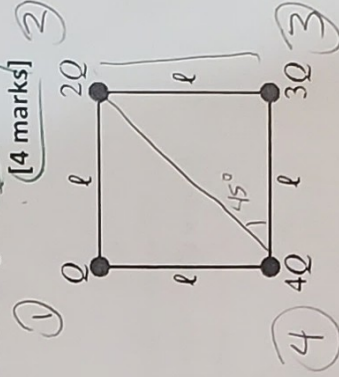
$$\vec{E} = kQ \left(\frac{1}{(x+a)^2} - \frac{1}{(x-a)^2} \right)$$

Final answer: $kQ \left(\frac{4ax}{(x+a)^2(x-a)^2} \right)$

directed left

$$(x^2 - a^2)^2$$

4. At each corner of a square of side ℓ there are point charges of magnitude Q , $2Q$, $3Q$, and $4Q$ as shown in the diagram. Determine the magnitude and direction of the force on the charge $2Q$.



$$\vec{F}_{1on2} = k \frac{Q \cdot 2Q}{\ell^2} = k \frac{Q^2}{\ell^2} (\hat{i})$$

$$\vec{F}_{3on2} = k \frac{2Q \cdot 3Q}{\ell^2} = k \frac{Q^2}{\ell^2} (\hat{j})$$

$$\vec{F}_{4on2} = (5)(45) \cdot \frac{kQ^2}{\ell^2} \left(\frac{8}{2} \right) = (2\sqrt{2}) \frac{kQ^2}{\ell^2}$$

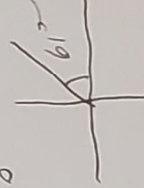
$$\sum \vec{F}_x = k \frac{Q^2}{\ell^2} (2) + 2\sqrt{2} \left(\frac{kQ^2}{\ell^2} \right)$$

$$\sum \vec{F}_x = k \frac{Q^2}{\ell^2} (2 + 2\sqrt{2})$$

$$\sum \vec{F}_y = k \frac{Q^2}{\ell^2} (6 + 2\sqrt{2})$$

$$\theta = \tan^{-1} \left(\frac{6 + 2\sqrt{2}}{2 + 2\sqrt{2}} \right)$$

$$\theta = 61.32^\circ$$



$$|\sum \vec{F}| = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$$

$$= \left(\frac{kQ^2}{\ell^2} \right)^2 \left((2 + 2\sqrt{2})^2 + (6 + 2\sqrt{2})^2 \right)$$

$$= \frac{kQ^2}{\ell^2} \sqrt{2343 + 77.91}$$

$$|\sum \vec{F}| = \frac{kQ^2}{\ell^2} (10.0625)$$

Final answer: $k \frac{Q^2}{\ell^2} (10.06) N$

61.32° North of east

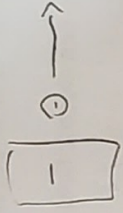
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5. An electron acquires 6.45×10^{-16} J of kinetic energy when it is accelerated by an electric field from plate A to plate B. What is the potential difference between the plates, and which plate is at the higher potential?

$$\frac{1}{2} m v^2 = 6.45 \times 10^{-16} = K E$$

[3 marks]



$$\frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 = - (E P B_f - E P E_f)$$

$$\Delta K E = - (V_f - V_i) \cdot q$$

$$\frac{\Delta K E}{-q} = \Delta V$$

$$4031.25 = \Delta V$$

Final answer: 4031.25 V

Thus the right plate has a potential that is 4031.25 V higher than the left.

6. A 35×10^{-6} C point charge is placed 46 cm from an identical 35×10^{-6} C charge. How much work would be required to move a 0.50×10^{-6} C test charge from a point midway between them to a point 12 cm closer to either of the charges? Show all your work.

[4 marks]

See solutions - $\Delta E P E$

$$W_{AB} = -q \Delta V = -q \left(\frac{1}{4\pi\epsilon_0} \left(\frac{1}{r_f} - \frac{1}{r_i} \right) \right)$$

$$E P I = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{(35 \times 10^{-6})^2}{(0.46)^2}$$

$$E P I = 8.69 \text{ kN}$$

$$W_{AB} = -\Delta E P E$$

$$= -\Delta V \cdot q$$

$$= - (11.27 \text{ kV} - 8.69 \text{ kV}) \cdot q$$

$$W_{AB} = - (2.57 \cdot 10^3) \cdot q$$

$$W_{AB} = - (2.57) (35 \times 10^{-6}) (0.5 \times 10^{-6}) (8.99 \times 10^9)$$

$$W_{AB} = 0.405 \text{ J}$$

← 0 F · d
so it's positive work done

Final answer: 0.405 J

COMMUNICATION

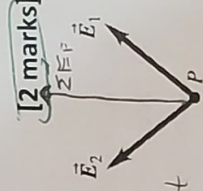
1. The electric field has a unique direction at any particular point, so only one field line can pass through each point of the field. In other words, field lines never intersect or cross each other.

- a) Why can electric field lines never cross?

That would imply that there are two fields at one spot which is not true since there can only be one net sum of fields at one point. So this means only one tangent can exist for that point, so only one field line ^{on that tangent}.

- b) The electric fields at point P due to the positive charges q_1 and q_2 are shown in the diagram. Does the fact that E_1 and E_2 cross each other violate the statement that electric field lines never cross? Explain.

This is because the vectors have to be added together using the principle of superposition. $\uparrow$ so there is only one $\Sigma \vec{E}$ at



meaning only one field line can be tangent. q_1 q_2

2. Describe clearly the difference between the following:

- a) The electric potential difference and electric field.

The electric potential difference is how much ^{need to mention vector vs scalar} work it takes to move a charge against the electric field.

- b) The electric potential difference and electric potential energy.

One is the potential in joules and the electric potential is the potential gained per coulomb of charge being moved. Need to ^{show more} show more.

Communication marks are awarded throughout the test according to the shown rubric.

[2 marks]

	Level 1	Level 2	Level 3	Level 4
Correct use of mathematical/physical symbols, labels, and conventions (e.g. units, proper component form for the values, positive direction given, etc...)	Few mathematical/physical conventions are used correctly. [1/2]	Some mathematical/physical conventions are used correctly. [1]	Most mathematical/physical conventions are used correctly. [1-1/2]	Consistently uses mathematical/physical conventions correctly. [2]